Kevin Jacob

ksjacob27@gmail.com | 720-419-7450 | github.com/ksjacob27 | linkedin.com/in/kevin-jacob-010595264

EDUCATION

University of Colorado Denver May 2026
Master of Science, Computer Science
University of Colorado Boulder May 2024
Bachelor of Science, Computer Science

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, TypeScript, JavaScript, SQL, Scala, C
Frameworks & Libraries: Spring Boot, FastAPI, Flask, Django, React, Node.js, Axios
AI/ML: OpenAI API, LangChain, LangGraph, AutoGen, Agno, Exa, RAG pipelines, Multi-agentic pipelines, prompt engineering, agentic systems, vector databases, PyTorch, TensorFlow, HuggingFace, Scikit-learn
Data & Cloud: Pandas, NumPy, MongoDB, MySQL, PostgreSQL, Cassandra, AWS Lambda, S3, SageMaker, Docker
Testing & Tools: Git, CI/CD, Agile, Jira, Postman, pytest, Mocha, Chai, Jupyter Notebook, Linux, Bash
Databases: MongoDB, MySQL, Cassandra, PostgreSQL, SQL
Certifications: AWS Certified AI Practitioner (2025), Software Engineering Graduate Certificate (CU Denver)

EXPERIENCE

Emergetech Business Solutions Inc. June 2024 - August 2024
Software Developer Intern

- Reduced average agent handle time by 18% by integrating the OpenAI API into a call center platform to generate AI-powered real-time response scripts, automating prompt-driven assistance across a full-stack Spring Boot and React application serving 200+ daily users.
- Cut manual reporting time by 40% by designing and building a dynamic SQL query builder that extracted, filtered, and aggregated records across large datasets using tag-based logic, eliminating ad-hoc querying for a team of 12 managers.
- Improved API reliability by 30% by engineering and testing 15+ REST endpoints with Postman and integrating Axios for fault-tolerant front-end/back-end communication, reducing integration failures caught in production.

Expedia Group August 2023 - April 2024
Software Engineering Capstone Member

- Delivered automated listing quality scores for 500+ VRBO properties by building an end-to-end AI-powered recommendation platform, reducing manual property review time by 65% by combining web scraping, LangChain orchestration, and OpenAI API inference deployed on AWS Lambda.
- Improved model output consistency by 45% by engineering and iteratively optimizing GPT prompts through systematic prompt regression testing, reducing malformed or inconsistent scoring responses across the LangChain pipeline.

OptIn Inc. June 2023 - August 2023
Applications Developer Intern

- Reduced scheduling conflicts by 70% across an employee portal serving 300+ workers by building a constraint-based predictive scheduling engine in Python and Django using Google OR-Tools, automating shift assignments by modeling labor demands, worker availability, and employee preferences.
- Achieved 98% test coverage across the scheduling pipeline by leading end-to-end unit and functional test development, catching 23 critical edge case failures before production deployment and reducing post-launch bug reports by 80%.

University of Colorado January 2023 - Present
Teaching Assistant
Algorithms, Database Systems, Intro to Programming

- Mentored 120+ students per semester across three courses by conducting weekly office hours, accelerating concept retention by debugging code in real time and translating complex algorithms, OOP design principles, and database systems into accessible explanations.

PROJECTS

F1 Pit Stop Strategy Optimizer | *Python, PyTorch, GRU, Scikit-learn, FastF1 API*

- Achieved 52.5% pit stop strategy accuracy across 122 race-driver combinations from the 2024 F1 season by designing a compound-specific GRU deep learning system trained on 28,000+ lap sequences from 2022-2023 race data, representing a +21.5% improvement over the brute-force baseline.
- Improved tire degradation prediction accuracy by 40% over the moving average baseline by engineering a lap-to-lap delta prediction target, building a full data pipeline on the FastF1 API with per-race MinMax normalization, and conducting five ablation experiments across nine model configurations.

AI Cybersecurity Defender | *Python, FastAPI, GPT-4o, LangChain, CICIDS2017*

- Achieved 94% network threat detection accuracy by designing a production-grade agentic AI system using GPT-4o and FastAPI, combining a rule-based classification engine with an LLM-driven reasoning pipeline to detect intrusions in real time across 28,000+ network samples in the CICIDS2017 benchmark dataset.

Lyric Analyzer | *Python, Scikit-learn, TensorFlow, HuggingFace, Spotify API*

- Reached 87% genre classification accuracy by building a full ML pipeline from data ingestion to deployment, training an SVM classifier with TF-IDF feature extraction on 10,000+ songs from the Spotify API and fine-tuning a feedforward neural network for generative lyric extension across five genre-specific corpora.

RELEVANT COURSEWORK

Machine Learning, Natural Language Processing, Algorithms, Database Systems, Operating Systems, Artificial Intelligence, Advanced Data Science, Cybersecurity, Object-Oriented Analysis and Design, Human Computer Interaction